after a glancing collision with a wayward Mars-sized protoplanet. The orbiting debris from this encounter coalesced to form our lovely, low-density satellite. Apart from this newsworthy event, the period of heavy bombardment that Earth endured during its infancy was not unique among the planets and other large bodies of the solar system. They each sustained similar damage, with the airless, erosionless surfaces of the Moon and Mercury preserving much of the cratered record from this period.

Not only is the solar system scarred by the flotsam of its formation, but nearby interplanetary space also contains rocks of all sizes that were jettisoned from Mars, the Moon, and Earth by the ground's recoil from high-speed impacts. Computer studies of meteor strikes demonstrate conclusively that surface rocks near impact zones can get thrust upward with enough speed to escape the body's gravitational tether. At the rate we are discovering meteorites on Earth whose origin is Mars, we conclude that about a thousand tons of Martian rocks rain down on Earth each year. Perhaps the same amount reaches Earth from the Moon. In retrospect, we didn't have to go to the Moon to retrieve Moon rocks. Plenty come to us, although they were not of our choosing and we didn't yet know it during the Apollo program.

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Most of the solar system's asteroids live and work in the main asteroid belt, a roughly flat zone between the orbits of Mars and Jupiter. By tradition, the discoverers get to name their asteroids whatever they like. Often drawn by artists as a region of cluttered, meandering rocks in the plane of the solar system, the asteroid belt's total mass is less than five percent that of the Moon, which is itself barely more than one percent of Earth's mass. Sounds insignificant. But accumulated perturbations of their orbits continually create a deadly subset, perhaps a few thousand, whose eccentric paths intersect Earth's orbit. A simple calculation reveals that most of them will hit Earth within a hundred million years. The ones larger than about a kilometer across will collide with enough energy to destabilize Earth's ecosystem and put most of Earth's land species at risk of extinction.

That would be bad.

Asteroids are not the only space objects that pose a risk to life on Earth. The Kuiper belt is a comet-strewn swath of circular real estate that begins just beyond the orbit of Neptune, includes Pluto, and extends